

ESC384 — Partial Differential Equations

Course Notes

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Unit 1:

Lecture 1: Introductions to PDEs

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Definition: A PDE is an equation that relates an unknown function u and some of its partial derivatives.

$$G\left(\underbrace{x_1, \dots, x_n}_{\text{spatial coordinates}}, \underbrace{t}_{\text{time}}, \underbrace{u}_{\text{function}}, \underbrace{\frac{\partial u}{\partial x_1}, \dots, \frac{\partial u}{\partial x_n}}_{\text{1st derivatives in space}}, \underbrace{\frac{\partial u}{\partial t}}_{\text{1st derivatives in time}}, \underbrace{\frac{\partial^2 u}{\partial x_1^2}, \frac{\partial^2 u}{\partial x_1 \partial x_2}, \dots}_{\text{2nd derivatives}}\right) = 0$$

Example: Model PDEs for this course.

Laplace's Equation:

$$-\Delta u = 0$$

$$\text{Laplacian: } -\nabla \cdot \nabla = -\frac{\partial^2}{\partial x_1^2} - \frac{\partial^2}{\partial x_2^2} \text{ in } \mathbb{R}^2$$

Physics: potential flow, electrostatics, steady heat transfer...

Poisson's Equation:

$$-\Delta u = f \longleftarrow \text{source function: e.g. heat}$$

Heat Equation:

$$\frac{\partial u}{\partial t} - \Delta u = f \quad \text{physics: unsteady heat transfer, diffusion of chemicals...}$$

Transport Equation:

$$\frac{\partial u}{\partial t} + \underbrace{b}_{\text{advection field}} \cdot \nabla u = f \quad \text{physics: transport of conserved quantities}$$

Wave Equation:

$$\frac{\partial^2 u}{\partial t^2} - \underbrace{c^2}_{c: \text{wave speed}} \Delta u = 0 \quad \text{physics: acoustics, vibration...}$$

Burger's Equation:

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = 0 \quad \text{physics: nonlinear waves...}$$

Classification of PDEs

Definition: The **order** of a PDE is the order of the highest derivative.

Example:

$$\begin{aligned}\text{Laplace's:} \quad & -\frac{\partial^2 u}{\partial x_1^2} - \frac{\partial^2 u}{\partial x_2^2} = 0 \quad \rightarrow \text{2nd order} \\ \text{Transport:} \quad & \frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} = 0 \quad \rightarrow \text{1st order} \\ \text{Heat:} \quad & \frac{\partial u}{\partial t} - \Delta u = 0 \quad \rightarrow \text{2nd order (1st in time, 2nd in space)}\end{aligned}$$

Definition: An operator $\mathcal{L}$ is a **linear operator** if

$$\mathcal{L}(\alpha w + \beta v) = \alpha \mathcal{L}w + \beta \mathcal{L}v$$

for any functions w & v and scalars α & β

Definition: A PDE is **linear** if it can be expressed as

$$\underbrace{\mathcal{L}}_{\text{linear operator}} u = \underbrace{f}_{\text{function independent of } u}$$

If PDE is not linear, then it is a nonlinear PDE.

Definition: A **linear homogeneous PDE** is a PDE of the form

$$\mathcal{L}u = 0$$

If a linear PDE is not homogeneous, then it is a linear nonhomogeneous PDE.

Example:

$$\begin{aligned}\text{Laplace's:} \quad & -\Delta u = 0 \quad \rightarrow \text{linear homogeneous} \\ \text{Poisson's:} \quad & -\Delta u = f, f \neq 0 \quad \rightarrow \text{linear nonhomogeneous} \\ \text{Heat:} \quad & \frac{\partial u}{\partial t} - \Delta u = 0 \quad \rightarrow \text{linear homogeneous} \\ \text{Burger's:} \quad & \frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = 0 \quad \rightarrow \text{nonlinear}\end{aligned}$$

Superposition

- If $\mathcal{L}u_1 = 0$ and $\mathcal{L}u_2 = 0$, then $\mathcal{L}(u_1 + u_2) = 0$
- In addition, if $\mathcal{L}u_0 = f$, then $\mathcal{L}(u_0 + u_1 + u_2) = f$

$\Rightarrow$ used frequently to solve linear PDEs

$\Rightarrow$ an unique solution found by imposing boundary (and initial) conditions

Classification of Second-Order PDEs

- Consider a general linear operator of the form

$$\mathcal{L}u = \sum_{i,j=1}^n a_{ij} \frac{\partial^2 u}{\partial x_i \partial x_j} + \sum_{i=1}^n b_i \frac{\partial u}{\partial x_i} + cu$$

(if problem is time-dependent, treat t as 0-th coordinate)

- Let $A \in \mathbb{R}^{n \times n}$ be a symmetric matrix s.t. $A_{ij} = a_{ij}$
- Let $\lambda_1, \dots, \lambda_n$ be the eigenvalues of A .
- Classifications:
 - **elliptic**: all eigenvalues are nonzero and have same sign
 - **hyperbolic**: all eigenvalues are nonzero and exactly one of them have a different sign
 - **parabolic**: exactly one eigenvalue is 0 and all others have the same sign

Example:

Laplace's: $\mathcal{L} = -\Delta = -\frac{\partial^2}{\partial x_1^2} - \frac{\partial^2}{\partial x_2^2}$ in $\mathbb{R}^2$

$$A = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \Rightarrow \lambda_1 = \lambda_2 = -1 \Rightarrow \text{elliptic}$$

Poisson's: elliptic

Wave: $\mathcal{L} = \frac{\partial^2}{\partial t^2} - \Delta = \frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial x_1^2} - \frac{\partial^2}{\partial x_2^2}$ (in $\mathbb{R}^2 \times \mathbb{R}$)

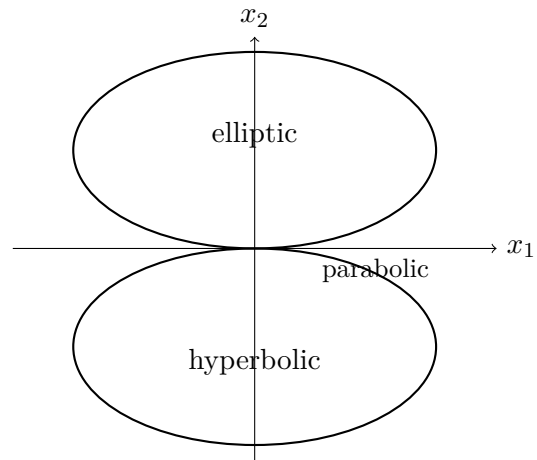
$$A = \begin{pmatrix} 1 & & \\ & -1 & \\ & & -1 \end{pmatrix} \Rightarrow \lambda_1 = 1, \lambda_2 = \lambda_3 = -1 \Rightarrow \text{hyperbolic}$$

Heat: $\mathcal{L} = \frac{\partial}{\partial t} - \Delta = \frac{\partial}{\partial t} - \frac{\partial^2}{\partial x_1^2} - \frac{\partial^2}{\partial x_2^2}$ in $\mathbb{R}^2 \times \mathbb{R}$

$$A = \begin{pmatrix} 0 & & \\ & -1 & \\ & & -1 \end{pmatrix} \Rightarrow \lambda_1 = 0, \lambda_2 = \lambda_3 = -1 \Rightarrow \text{parabolic}$$

Variable Coefficient

$$x_2 \frac{\partial^2 u}{\partial x_1^2} + \frac{\partial^2 u}{\partial x_2^2} = 0 \Rightarrow A = \begin{pmatrix} x_2 & 0 \\ 0 & 1 \end{pmatrix} \Rightarrow \lambda_1 = x_2, \lambda_2 = 1$$



Lecture 2: Fourier Series

September 14, 2026

Motivation: to solve PDEs, we must represent solution u as a function of space.

Given some function:

Approach 1: describe point-wise values

Approach 2: linear combination of known functions (e.g. polynomial)

$$u(x) = \sum_{n=1}^{\infty} \hat{u}_n \varphi_n(x)$$

$\Rightarrow$ can be easily differentiated: $u'(x) = \sum_{n=1}^{\infty} \hat{u}_n \varphi'_n(x) \Rightarrow$ classical method Fourier Series (sin & cos)

Definition: $f : \mathbb{R} \rightarrow \mathbb{R}$ is z -periodic if

$$f(x+z) = f(x) \forall x \in \mathbb{R}$$

Fundamental Period: smallest z for which the relationship holds

Example:

$$f(x) = \sin(x) \Rightarrow 2\pi \text{ periodic}$$

$$f(x) = \sin(x) + \cos(2x) \Rightarrow 2\pi \text{ periodic}$$

Definition: $f : \mathbb{R} \rightarrow \mathbb{R}$ is even if $f(-x) = f(x)$, sym about y $f : \mathbb{R} \rightarrow \mathbb{R}$ is odd if $f(-x) = -f(x)$, sym about origin.

Definition: Given $f : (0, a) \rightarrow \mathbb{R}$, its periodic extension to $\mathbb{R}$ is $f^p : \mathbb{R} \rightarrow \mathbb{R}$ such that $f^p(x) = f(x), x \in (0, a)$ and has a period of " a ".

Definition: Given $f : (0, a) \rightarrow \mathbb{R}$, its even periodic extension to $\mathbb{R}$ is $f^{e,p} : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f^{e,p}(x) = \begin{cases} f(x), & x \in (0, a) \\ f(-x), & x \in (-a, 0) \end{cases} \quad \text{and has a period of } 2a$$

Definition: Given $f : (0, a) \rightarrow \mathbb{R}$, its odd periodic extension to $\mathbb{R}$ is $f^{o,p} : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f^{o,p}(x) = \begin{cases} f(x), & x \in (0, a) \\ -f(-x), & x \in (-a, 0) \end{cases} \quad \text{and has a period of } 2a$$

Definition: Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a 2-periodic function, the **Fourier Series** of f is

$$f(x) \sim \frac{1}{2}a_0 + \sum_{n=1}^{\infty} (a_n \cos(n\pi x) + b_n \sin(n\pi x))$$

with Fourier Coefficients

$$a_n = \int_{-1}^1 \cos(n\pi x) f(x) dx, \quad n = 0, 1, 2, \dots$$

$$b_n = \int_{-1}^1 \sin(n\pi x) f(x) dx, \quad n = 1, 2, 3, \dots$$

Definition: Truncated Fourier Series is the partial sum

$$S_N(x) = \frac{1}{2}a_0 + \sum_{n=1}^N (a_n \cos(n\pi x) + b_n \sin(n\pi x))$$

Observations

1. In general, the Fourier series of f is not equal to f , and hence we use " " instead of "=".
2. The set of functions

$$\left\{ \frac{1}{\sqrt{2}}, \sin(\pi x), \cos(\pi x), \sin(2\pi x), \cos(2\pi x), \dots \right\}$$

is orthonormal with respect to the L^2 inner product

$$(w, v) = \int_{-1}^1 wv dx \quad w, v \in \mathbb{R}^2, w \cdot v = 0$$

For all $n, m = 1, 2, 3 \dots$

- $\int_{-1}^1 \left(\frac{1}{\sqrt{2}}\right) (\cos(m\pi x)) dx = \frac{1}{\sqrt{2}m\pi} \sin(m\pi x) \Big|_{x=-1}^1 = 0$
 - $\int_{-1}^1 \sin(m\pi x) dx = 0$
 - $\int_{-1}^1 \cos(n\pi x) \cos(m\pi x) dx - \delta_{mn} = \begin{cases} 1 & \text{if } m = n \\ 0 & \text{otherwise} \end{cases}$
 - $\int_{-1}^1 \sin(n\pi x) \sin(m\pi x) dx = \delta_{mn}$
 - $\int_{-1}^1 \sin(n\pi x) \cos(m\pi x) dx = 0$
3. Expression for Fourier coefficients are a consequence of orthogonality e.g. consider a_2
want: $f(x) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty} (a_n \cos(n\pi x) + b_n \sin(n\pi x))$
 multiply by $\cos(2\pi x)$ and integrate over $(-1, 1)$

$$\begin{aligned} \int_{-1}^1 \cos(2\pi x) f(x) dx &= \int_{-1}^1 \cos(2\pi x) \left[\frac{1}{2}a_0 + \sum_{n=1}^{\infty} (a_n \cos(n\pi x) + b_n \sin(n\pi x)) \right] dx \\ &= a_0 \underbrace{\int_{-1}^1 \frac{1}{2} \cos(2\pi x) dx}_0 + \sum_{n=1}^{\infty} a_n \underbrace{\int_{-1}^1 \cos(2\pi x) \cos(n\pi x) dx}_{S_{n2}} + \sum_{n=1}^{\infty} \underbrace{\int_{-1}^1 \cos(2\pi x) \sin(n\pi x) dx}_{=0} \\ &= \sum_{n=1}^{\infty} a_n S_{n2} = a_2 \end{aligned}$$

4. If $f : (-1, 1) \rightarrow \mathbb{R}$, then its Fourier series is "equal to"
- (i) f as $(-1, 1)$

(ii) its periodic extension f^p in $\mathbb{R}$

Example:

$$f(x) = x \text{ for } x \in (-1, 1)$$

$$\text{want: } f(x) \sim \frac{1}{2}a_0 + \sum_{n=1}^{\infty} (a_n \cos(n\pi x) + b_n \sin(n\pi x))$$

$$a_n = \int_{-1}^1 f(x) \cos(n\pi x) dx = \int_{-1}^1 x \cos(n\pi x) dx = 0$$

$$\begin{aligned} b_n &= \int_{-1}^1 x \sin(n\pi x) dx = \left[\frac{\sin(n\pi x)}{n^2 \pi^2} \right]_{x=-1}^1 - \left[\frac{x \cos(n\pi x)}{n\pi} \right]_{x=-1}^1 \\ &= -1 = -2 \frac{\cos(n\pi)}{n\pi} = -\frac{2(-1)^n}{n\pi} = 2 \frac{(-1)^{n+1}}{n\pi} \end{aligned}$$

Hence,

$$f(x) \sim \sum_{n=1}^{\infty} b_n \sin(n\pi x) = \boxed{\sum_{n=1}^{\infty} 2 \frac{(-1)^{n+1}}{n\pi} \sin(n\pi x)}$$

Definition: Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be an even z -periodic function. The **Fourier cosine series** is

$$f(x) \sim \frac{1}{2} \hat{f}_0 + \sum_{n=1}^{\infty} \hat{f}_n \cos(n\pi x)$$

$$\text{where } \hat{f}_n = \int_{-1}^1 \cos(n\pi x) f(x) dx = 2 \int_0^1 \cos(n\pi x) f(x) dx$$

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be an even z -periodic function. The **Fourier sine series** is

$$f(x) \sim \sum_{n=1}^{\infty} \hat{f}_n \sin(n\pi x)$$

$$\text{where } \hat{f}_n = \int_{-1}^1 \sin(n\pi x) f(x) dx = 2 \int_0^1 \sin(n\pi x) f(x) dx$$

Observations:

1. If $f : (0, 1) \rightarrow \mathbb{R}$, then its Fourier cosine/sine series is "equal to"
 - (i) f as $(0, 1)$ & its even/odd periodic extension as $\mathbb{R}$
2. We use $\hat{f}_n$ to denote Fourier cosine/sine coefficients of f

Example: Fourier cosine series of $f(x) = x, x \in (0, 1)$

$$\hat{f}_0 = 2 \int_0^1 x dx = 1$$

$$\hat{f}_n = 2 \int_0^1 \cos(n\pi x) x dx = \begin{cases} -\frac{4}{n^2 \pi^2}, & n \text{ odd} \\ 0, & n \text{ even} \end{cases}$$

$$\begin{aligned} f(x) &\sim \frac{1}{2} \hat{f}_0 + \sum_{n=1}^{\infty} \hat{f}_n \cos(n\pi x) = \frac{1}{2} + \sum_{n=1, n \in \text{odd}}^{\infty} \left(-\frac{4}{n^2 \pi^2}\right) \cos(n\pi x) \\ &= \frac{1}{2} + \sum_{k=1}^{\infty} -\frac{4}{(2k-1)^2 \pi^2} \cos((2k-1)\pi x) \end{aligned}$$

Example: Fourier sine series of $f(x) = x, x \in (0, 1)$

$$\hat{f}_n = 2 \int_0^1 \sin(n\pi x) dx = \dots = \frac{2}{n\pi}(-1)^{n_1}$$

$$f(x) \sim \sum_{n=1}^{\infty} \hat{f}_n \sin(n\pi x) = \sum_{n=1}^{\infty} \frac{2(-1)^{(n+1)}}{n\pi} \sin(n\pi x)$$

Obstructions:

1. Coefficients follow from orthogonality

$$\frac{1}{L} \int_{-L}^L \sin\left(\frac{n\pi x}{L}\right) \sin\left(\frac{m\pi x}{L}\right) dx = \delta_{mn}, \quad \text{etc}$$

2. Given $f : (-L, L) \rightarrow \mathbb{R}$, its Fourier series converges to $2L$ - periodic extension on $\mathbb{R}$.

Lecture 3: Fourier Series: Analysis

Sep 18, 2026

$$f(x) \sim \frac{1}{2}a_0 + \sum_{n=1}^{\infty} (a_n \cos(n\pi x) + b_n \sin(n\pi x))$$

$$S_N(x) = \frac{1}{2}a_0 + \sum_{n=1}^N (a_n \cos(n\pi x) + b_n \sin(n\pi x))$$

Question: Does $S_N \rightarrow f$ as $N \rightarrow \infty$? In what sense? For what functions?

Regularity of Functions

Definition: $f : (a, b) \rightarrow \mathbb{R}$ is continuous on (a, b) if $\lim_{x \rightarrow c} f(x) = f(c)$, $\forall c \in (a, b)$

Definition: $f : [a, b] \rightarrow \mathbb{R}$ is continuous on $[a, b]$ if $\lim_{x \rightarrow c} f(x) = f(c)$, $\forall c \in (a, b)$ and

$$\lim_{x \rightarrow a^+} f(x) = f(a) \quad \text{and} \quad \lim_{x \rightarrow b^-} f(x) = f(b)$$

Definition: $f : I \rightarrow \mathbb{R}$ is continuously differentiable on I if f' is continuous on I .

Example:

$$f(x) = |x|, \quad x \in (-1, 1)$$

Example:

$$f(x) = \sqrt{x}, \quad x \in [0, 1]$$

Definition: $f : (a, b) \rightarrow \mathbb{R}$ is piecewise continuous on (a, b)

if $\exists$ a set of points $a \equiv x_1 < x_2 < \dots < x_N \equiv b$ s.t. f is continuous on each (x_i, x_{i+1}) and one-sided limits $\lim_{x \rightarrow x_i^+} f(x)$ and $\lim_{x \rightarrow x_{i+1}^-} f(x)$ exist.

Definition: $f : (a, b) \rightarrow \mathbb{R}$ is piecewise continuously differentiable on (a, b) if f' is piecewise continuous.

Convergence of Fourier Series

Theorem Pointwise Convergence. Let f be a piecewise continuously differentiable function with a period 2. Then, for any $x \in \mathbb{R}$

$$S_N(x) \rightarrow \frac{1}{2}[f(x^-) + f(x^+)] \text{ as } N \rightarrow \infty$$

Corollary. if f is continuous at x , then $S_N(x) \rightarrow f(x)$

if f is not continuous at x , then $S_N(x) \rightarrow$ average of one-sided limits.

Theorem Uniform Convergence. Let f be a continuous and piecewise continuously differentiable function with a period 2. Then

$$\mathcal{E}_N = \max_{x \in \mathbb{R}} |f(x) - S_N(x)| \rightarrow 0 \text{ as } N \rightarrow \infty$$

If a series is uniformly convergent

1. S_N gives a good approximation to f at all points (to within $\mathcal{E}_N$)
2. We can make $\mathcal{E}_N$ arbitrarily small by taking N large
3. f must be continuous (cf. uniform limit thm)
4. Fourier series is equal to f

$$f(x) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty} (a_n \cos(n\pi x) + b_n \sin(n\pi x))$$

Theorem L^2 Convergence. Let f be a square integrable function with a period 2: i.e.

$$\int_{-1}^1 f(x)^2 dx < \infty$$

then,

$$\int_{-1}^1 (f(x) - S_N(x))^2 dx \rightarrow 0 \text{ as } N \rightarrow \infty$$

Example:

$$f(x) = \text{sgn}(\sin(\pi x))$$

Pointwise convergent? $\rightarrow$ piecewise continuously differentiable? YES

Uniform convergent? $\rightarrow$ continuous? NO

Example:

$$f(x) = |x|$$

Convergence on Finite Intervals

- Fourier series converges pointwise/uniformly/ L^2 if the appropriate periodic extension satisfies the required continuity & differentiability conditions.
- Caveat: even if $f : [-L, L] \rightarrow \mathbb{R}$ is continuous, periodic extension may not be continuous

Endpoint Conditions for Continuity

- full: $f : [-L, L] \rightarrow \mathbb{R}$, $f(-L^+) = f(L^-)$
- sine: $f : [0, L] \rightarrow \mathbb{R}$, $f(0^+) = f(L^-) = 0$
- cosine: $f : [0, L] \rightarrow \mathbb{R}$, no additional requirements.
- "Jump" (if any) at endpoints determines the value of Fourier series @ endpoints.

Uniform and Absolute Convergence

Theorem. Fourier series $f(x) \sim \frac{1}{2}a_0 + \sum_{n=1}^{\infty}(a_n \cos(n\pi x) + b_n \sin(n\pi x))$ converges uniformly if

$$\sum_{n=1}^{\infty}(|a_n| + |b_n|) < \infty$$

Example:

$$f(x) = x, x \in (0, 1)$$

Fourier cosine series: $\hat{f}_n = \frac{2}{n^2\pi^2}((-1)^n - 1)$

$$\sum_{n=1}^{\infty} |\hat{f}_n| \leq \sum_{n=1}^{\infty} \frac{4}{n^2\pi^2} < \infty \Rightarrow \text{uniform conv.}$$

Fourier sine series: $\hat{f}_n = \frac{2}{n\pi}(-1)^{n+1}$

$$\sum_{n=1}^{\infty} |\hat{f}_n| = \sum_{n=1}^{\infty} \frac{2}{n\pi} \Rightarrow \text{diverges} \Rightarrow \text{may not uniformly converge}$$

Differentiation of Fourier Series

Theorem. Let f be a continuous periodic function with piecewise continuously differentiable f' . Then, the Fourier series of f' is given by term-by-term differentiation:

$$f(x) \sim \frac{1}{2}a_0 + \sum_{n=1}^{\infty}(a_n \cos(n\pi x) + b_n \sin(n\pi x))$$

$$f'(x) \sim \sum_{n=1}^{\infty}(-a_n n\pi \sin(n\pi x) + b_n n\pi \cos(n\pi x))$$

Theorem Uniform Convergence of Derivatives. The k -th derivative of f is continuous if

$$\sum_{n=1}^{\infty}(|n^k a_n| + |n^k b_n|) < \infty$$

Example:

$$f(x) = \sum_{n=1}^{\infty} \underbrace{\exp(-\lambda n)}_{b_n} \sin(n\pi x) \text{ for } \lambda > 0$$

$$\sum_{n=1}^{\infty} n^k |b_n| = \sum_{n=1}^{\infty} n^k \exp(-\lambda n) < \infty \text{ for any } k$$

Lecture 4: Sturn-Liouville Problems

09.21

Fourier Series & Eigenproblems

Example: Find eigenfunctions ϕ_n and eigenvalues λ_n s.t.

$$\begin{cases} -\phi_n'' = \lambda_n \phi_n & \text{in } (0, 1) & \text{ODE} \\ \phi_n(x=0) = \phi_n(x=1) = 0 & & \text{BCs} \end{cases}$$

Case 1: $\lambda_n < 0$

Let $\mu_n = -\lambda_n$ so that $\mu_n > 0$

General solution to ODE: $\phi_n'' = \mu_n \phi_n$

$$\begin{aligned} \phi_n &= A_n' \exp(\sqrt{\mu_n}x) + B_n' \exp(-\sqrt{\mu_n}x) \\ &= A_n \cosh(\sqrt{\mu_n}x) + B_n \sinh(\sqrt{\mu_n}x) \end{aligned}$$

Recall: $\cosh(x) = \frac{1}{2}(e^x + e^{-x})$, $\sinh(x) = \frac{1}{2}(e^x - e^{-x})$

$$\phi_n''(x) = A_n \mu_n \cosh(\sqrt{\mu_n}x) + B_n \mu_n \sinh(\sqrt{\mu_n}x) = \mu_n \phi_n$$

BC $x=0$: $\phi_n(x=0) = A_n \cosh(0) + B_n \sinh(0) = A_n = 0 \Rightarrow A_n = 0$

BC $x=1$: $\phi_n(x=1) = B_n \sinh(1) = 0 \Rightarrow B_n = 0$

$\Rightarrow$ only solution is the trivial solution $\phi_n = 0$

Case 2: $\lambda_n = 0$

ODE: $-\phi_n'' = 0 \Rightarrow \phi_n(x) = A_n x + B_n$

BC $x=0$: $\phi_n(x=0) = B_n = 0 \Rightarrow B_n = 0$

BC $x=1$: $\phi_n(x=1) = A_n = 0 \Rightarrow A_n = 0$

Rightarrow only trivial solution $\phi_n = 0$

Case 3: $\lambda_n > 0$

ODE: $-\phi_n'' = \lambda_n \phi_n$

$$\Rightarrow \phi_n(x) = A_n \cos(\sqrt{\lambda_n}x) + B_n \sin(\sqrt{\lambda_n}x)$$

BC $x=0$: $\phi_n(x=0) = A_n \cos(0) + B_n \sin(0) = A_n = 0 \Rightarrow A_n = 0$

BC $x=1$: $\phi_n(x=1) = B_n \sin(\sqrt{\lambda_n} \cdot 1) = 0$

$\Rightarrow$ since we want nontrivial solution, take $B_n \neq 0$

$\Rightarrow \sin(\sqrt{\lambda_n}) = 0 \Rightarrow \sqrt{\lambda_n} = n\pi \Rightarrow \lambda_n = n^2\pi^2, n = 1, 2, 3, \dots$

$\Rightarrow \phi_n(x) = \sin(n\pi x)$ & $\lambda_n = n^2\pi^2$ are the (nontrivial) solutions of the eigenproblem.

Observations

1. Eigenfunctions for the basis of Fourier sine series. $\hookrightarrow$ they can be used to approximate functions
2. Eigenfunctions are orthogonal
3. Eigenvalues are real and non-negative Q. could we have known this just by looking at the eigenproblem?

Intuition: eigenpairs of symmetric matrices

Sturm-Liouville Theory

Definition: Sturm-Liouville problem is an eigenproblem of the form

$$\begin{cases} (p\phi')' + q\phi = \lambda w\phi & \text{in } (a, b) & \text{ODE} \\ \alpha_1\phi(a) - \alpha_2\phi'(a) = 0 & & \text{left BC} \\ \beta_1\phi(b) + \beta_2\phi'(b) = 0 & & \text{right BC} \end{cases}$$

where:

- (i) $p, p'mq, & w$ are continuous functions on $[a, b]$.
- (ii) $p > 0$ & $w > 0$ on $[a, b]$
- (iii) $\alpha_1^2 + \alpha_2^2 > 0$
- (iv) $\beta_1^2 + \beta_2^2 > 0$

Example:

$$\begin{cases} -\phi'' = \lambda\phi & \text{in } (0,1) \\ \phi(0) = \phi(1) = 0 \end{cases}$$

is Sturm-Liouville problem with $p = 1, q = 0; w = 1, \alpha_1 = 1, \alpha_2 = 0, \beta_1 = 1, \beta_2 = 2$

Theorem. Sturm-Liouville operator $L = -\frac{d}{dx}(p\frac{d}{dx}) + q$ is self-adjoint: i.e,

$$(Lw, v) = (w, Lv)$$

for any continuously twice differentiable functions w & v when $(f, g) = \int_a^b f\bar{g} dx$ for $\bar{g}$ is the complex conjugate.

Analogy: if A is symmetric, then

$$(Aw, v) - v^T Aw = w^T A^T v = w^T Av = (w, Av)$$

Theorem. All eigenvalues of S-L problems are real.

Analogy: all eigenvalues of symmetric A are real.

Theorem. Eigenfunctions of S-L problems associated with distinct eigenvalues λ_n & λ_m are orthogonal wrt weight w :

$$\int_a^b w\phi_n\phi_m dx = 0, \quad \forall n \neq m$$

Analogy: eigenvectors of symmetric A associated with distinct eigenvalues are orthogonal.

Theorem. S-L problem has an infinite number of eigenvalues and $\lambda_n \rightarrow \infty$ as $n \rightarrow \infty$.

Theorem. If $q \geq 0, \alpha_1 \geq 0, \alpha_2 \geq 0, \beta_1 \geq 0 \& \beta_2 \geq 0$,
then $\lambda_n \geq 0 \quad \forall n$

Key: given

$$\begin{cases} -\phi_n'' = \lambda_n \phi_n & \text{in } (0,1) \\ \phi_n(0) = \phi_n(1) = 0 \end{cases}$$

we could have known, without knowing the solutions

- i eigenfunctions are orthogonal
- ii eigenvalues are real & non-negative

Example:

$$\begin{cases} -\phi'' = \lambda \phi & \text{in } (0,1) \quad \text{ODE} \\ \phi'(0) = \phi'(1) = 0 & \text{BCs} \end{cases}$$

S-L problem with $p = 1, q = 0, w = 1, \alpha_1 = \beta_1 = 0, \alpha_2 = \beta_2 = 1$

Rightarrow eigenfunctions are orthogonal; eigenvalues are real & non-negative.

Explicit Solution:

Case 1: $\lambda = 0$ (note: we need not consider $\lambda < 0$ by S-L theory)

$$\text{ODE: } -\phi_n'' = 0 \Rightarrow \phi_n(x) = A_n(x) + B_n$$

$$\text{BC } x = 0 : \quad \phi_n'(0) = A_n = 0 \Rightarrow A_n = 0$$

$$\text{BC } x = 1 : \quad \phi_n'(1) = A_n = 0$$

$$\Rightarrow \text{pick } B_n = 1 \Rightarrow \phi_n(x = 1), \lambda_n = 0$$

Case 2: $\lambda_n > 0$

$$\text{ODE: } -\phi_n'' = \lambda_n \phi_n \Rightarrow \phi_n(x) = A_n \cos(\sqrt{\lambda_n} x) + B_n \sin(\sqrt{\lambda_n} x)$$

$$\text{BC } x = 0 : \quad \phi_n'(0) = -A_n \sqrt{\lambda_n} \sin(\sqrt{\lambda_n} \cdot 0) + B_n \sqrt{\lambda_n} \cos(\sqrt{\lambda_n} \cdot 0) = 0 \Rightarrow B_n = 0$$

$$\text{BC } x = 1 : \quad \phi_n'(1) = -A_n \sqrt{\lambda_n} \sin(\sqrt{\lambda_n} \cdot 1) = 0 \Rightarrow \sin(\sqrt{\lambda_n}) = 0 \Rightarrow \lambda_n = n^2 \pi^2, n = 1, 2, \dots$$

$$\phi_n(x) = \begin{cases} 1, & n=0 \\ \cos(n\pi x), & n=1, 2, 3, \dots \end{cases} = \cos(n\pi x), n = 0, 1, 2, \dots$$

$$\lambda_n = \begin{cases} 0, & n=0 \\ n^2 \pi^2, & n=1, 2, 3, \dots \end{cases} = n^2 \pi^2, n = 0, 1, 2, \dots \Rightarrow \text{Fourier cosine series}$$

Generalized Fourier Series

Let $(\phi_n)_{n=1}^\infty$ be eigenfunctions of the S-L problem with $\alpha_1, \alpha_2, \beta_1, \beta_2 \geq 0$.

We can approximate any square integrable $f : (a, b) \rightarrow \mathbb{R}$ by generalized Fourier series.

$$f(x) \sim \sum_{n=1}^{\infty} \hat{f}_n \phi_n(x)$$

where

$$\hat{f}_n = \frac{\int_a^b f(x) \phi_n(x) w(x) dx}{\int_a^b \phi_n(x)^2 w(x) dx}$$

note: the basis is orthogonal, not necessarily orthonormal.

Remarks

:

1. Coefficients follow from orthogonality
2. Series converges point-wise/uniformly/in L^2 given a sufficiently regular f .